DSA Experiment 8

Title: Implement Binary search tree with operations Create, search, and recursive traversal.

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CODE:

```
#include  <stdio.h>
#include <stdlib.h>
struct Node
{
    int data;
    struct Node *left;
    struct Node *right;
};

struct Node *createNode(int value)
{
    struct Node *newNode =
        (struct Node *)malloc(sizeof(struct Node));

    if (newNode == NULL)
        return NULL;

    newNode->data = value;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}

struct Node *insert(struct Node *root, int value)
{
    if (root == NULL)
        return createNode(value);

    if (value < root->data)
        root->left = insert(root->left, value);
    else if (value > root->data)
        root->right = insert(root->right, value);

    return root;
}
```

```
struct Node *search(struct Node *root, int key)
{
    if (root == NULL || root->data == key)
        return root;

    if (key < root->data)
        return search(root->left, key);

    return search(root->right, key);
}

int main()
{
    struct Node *root = NULL;
    struct Node *result;

    root = insert(root, 200);
    root = insert(root, 55);
    root = insert(root, 45);
    root = insert(root, 20);
    root = insert(root, 89);
    root = insert(root, 18);

    int key = 45;

    result = search(root, key);

    if (result != NULL)
        printf("Element found: %d\n", result->data);
    else
        printf("Element not found\n");

    return 0;
}
```

Output:

```
Output

Element found: 45
```